Paper I – Basic Sciences in Urology (MCh Urology, Summer 2018)

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I-1. Describe anatomy of lymphatic drainage of genitourinary system, etiopathogenesis and management of Chyluria

Type: LAQ (25 marks)

General plan of lymphatic drainage

- Lymph flows **cephalad and right → left**.
- Pelvic nodes (external iliac, internal iliac, obturator, presacral) → common iliac → lumbar (para-aortic, interaortocaval, paracaval) nodes → **lumbar trunks**. These join the intestinal trunk at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). The **thoracic duct** rises behind the aorta to the left innominate vein.
- Pelvic lymph crosses freely between right and left. Organs supplied by lateral aortic branches (kidney, adrenal, ureter, gonad) drain to lumbar nodes.

Drainage by organ

Organ	Primary nodes	Notes
Left kidney	Left lateral para-aortic	Capsular and deep hilar networks drain to renal vein nodes; may go retrocrural
Right kidney	Right paracaval, interaortocaval	May cross to left para-aortic
Ureter	Abdominal: para-aortic (left), paracaval/interaortocaval (right); middle: common iliac; pelvic: common, external, internal iliac	Plexus in muscle and adventitia
Bladder	Mainly **external iliac**; also internal iliac, obturator	Begins in lamina propria; bilateral cross-drainage
Prostate	**Obturator and internal iliac**; presacral	Three routes: superolateral (side wall), posterior (presacral), apical (internal pudendal)
Testis	Left: left para-aortic; right: interaortocaval, then right paracaval	Follows gonadal vessels; crossover only right → left. Epididymal cauda and vas → external iliac

Organ	Primary nodes	Notes
Penis	Skin → superficial inguinal (both groins); glans and corpora → superficial and deep inguinal, then pelvic	Shaft lymphatics converge dorsally
Scrotum	Ipsilateral superficial inguinal	Does not cross the raphe
Urethra (male)	Penile → inguinal; bulbomembranous and prostatic → internal/external iliac, obturator	Female: proximal → pelvic; distal → inguinal

Chyluria: definition

- Passage of **chyle** (intestinal lymph rich in chylomicrons) in urine, giving milky urine. Caused by an abnormal **lymphourinary (chylourinary) fistula**, usually at the renal fornices.

Etiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis)	Congenital lymphangiectasia, renal lymphangioma
	Trauma, surgery (e.g. partial nephrectomy, RPLND (retroperitoneal lymph node dissection))
	Retroperitoneal tumour or fibrosis, tuberculosis, pregnancy

Pathogenesis (obstructive theory)

- Adult worms live in retroperitoneal lymphatics and cause lymphangitis and fibrosis. This **obstructs** flow between the intestinal lymph trunks and the thoracic duct.
- Proximal lymphatics dilate and their valves fail, so chyle **refluxes** into renal hilar lymphatics.
- Raised pressure ruptures them into a calyceal fornix → **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- Usually intermittent; may remit spontaneously.

Clinical features

- Milky urine, worse after fatty meals; chylous clots (retention, colic); haematochyluria.
- Loss of fat and protein → weight loss, hypoproteinaemia, anaemia, lowered immunity.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Evaluation

- **Confirm chyle: ether (chloroform) test**: urine shaken with ether clears because fat dissolves. **Urine triglycerides** positive; Sudan III stains fat globules. Distinguish from phosphaturia (clears with acetic acid) and pyuria.
- **Filarial tests:** night blood smear (often negative at this stage), filarial antigen test, eosinophilia.
- **Lateralise the side: cystoscopy after a fatty meal** shows milky efflux from the ureteric orifice. Retrograde pyelography shows **pyelolymphatic backflow**.
- **Define fistula: lymphoscintigraphy** (non-invasive, shows tracer in kidney and bladder), pedal lymphangiography (historical gold standard), MR lymphangiography or CT.
- Serum albumin, haemoglobin, renal function.

Management

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Mild → diet + DEC
Persistent / moderate → add sclerotherapy (can repeat)
Severe or failed sclerotherapy → renal hilar lymphatic disconnection
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1. Conservative

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids. MCTs are absorbed straight into the portal blood, not into lacteals, so chyle formation falls.
- **DEC (diethylcarbamazine)** 6 mg/kg/day for 12 days kills microfilariae. Albendazole, ivermectin or doxycycline (anti-*Wolbachia*) are alternatives.

2. Endoscopic sclerotherapy (renal pelvic instillation)

- Ureteric catheter on the affected side, head-down. Instil **silver nitrate** (0.5–1%) or **povidone-iodine**, repeated over several sessions.
- Chemical lymphangitis and fibrosis seal the fistula. Good initial success; recurrence common.
- Complications: flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention.
- **Steps:** full renal mobilisation (nephrolysis) → **skeletonise the renal artery and vein**, ligating or sealing all hilar lymphatics → strip the upper ureter → remove perinephric fat → nephropexy ± omental wrap around the hilum.
- Reflux occurs only through hilar lymphatics, so upper-pole dissection can be limited. On-table cystoscopy confirms clear urine.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. Retroperitoneoscopic ligation beats open (shorter operation and stay, less blood loss, fewer complications).
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy obsolete; prevent by clipping lymphatics during renal and nodal surgery.

Diagram to draw: Retroperitoneal lymphatics with cisterna chyli, obstruction and pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Right → left lymph flow; cisterna chyli at L1–L2.
> - Chyluria = pyelolymphatic fistula, usually filarial (*W. bancrofti*).
> - Ether test and triglycerides confirm; cystoscopy after fatty meal lateralises.
> - Diet (MCT) + DEC → sclerotherapy → laparoscopic hilar lymphatic disconnection.

I-2. Discuss types of access, gases used in Laparoscopic urology and describe comparative analysis of open, Laparoscopic and Robotic Radical prostatectomy.

Type: LAQ (25 marks)

Access in laparoscopy

A. Transperitoneal: closed (Veress)

- Veress needle: 14-gauge, spring-loaded blunt stylet. Check flow and tip recoil first. Two "pops": fascia, then peritoneum.
- **Sites:** umbilicus (thinnest wall; aim towards pelvis in non-obese); lower quadrants; **Palmer point** (left subcostal, midclavicular line) with suspected adhesions; avoid scars.
- **Confirmation tests:** aspiration–injection–aspiration of saline; **drop test**; advancement test (1 cm free). Then insufflate: opening pressure <8–10 mmHg, set limit 15 mmHg.
- First trocar inserted blind, or under vision with an **optical trocar** (0° lens inside a clear tip).

B. Open (Hasson)

- 2-cm infra-umbilical incision; fascia and peritoneum opened under vision; stay sutures; blunt cannula.
- Safer with adhesions or previous surgery and early in training; more gas leak.

C. Retroperitoneal

- 2-cm incision below the tip of the 12th rib; finger dissection behind Gerota's fascia, in front of psoas → **balloon dilatation** (commercial or **Gaur** glove-on-catheter balloon) → balloon-tip port to prevent gas leak and emphysema.

D. Extraperitoneal (pelvic): infra-umbilical, behind rectus into the Retzius space with a balloon (used for extraperitoneal prostatectomy).

E. Others: hand-assisted (6.5–7.5 cm port); single-port/LESS (laparoendoscopic single-site); direct trocar entry without pneumoperitoneum.

- **Trocars: blunt dilating trocars** have replaced bladed ones (about fivefold less epigastric injury, fewer hernias). Close fascia of midline ports >10 mm or stretched sites (all ≥5 mm in children).
- **Access complications:** preperitoneal insufflation, vessel or bowel puncture, gas embolism, subcutaneous emphysema.

Gases used

Gas	Advantages	Disadvantages
CO_2 (carbon dioxide; standard)	Non-combustible, cheap, **highly blood-soluble** (embolism less lethal), quickly absorbed	Absorbed → hypercarbia, acidosis, arrhythmia; dangerous in COPD (chronic obstructive pulmonary disease)
Nitrous oxide	Less peritoneal pain, fewer acid–base effects	**Supports combustion**: no electrosurgery
Helium	Inert, no hypercarbia; use in lung disease	Poorly soluble → embolism; costly
Argon	Inert	Embolism risk
Air, oxygen	–	Embolism, combustion: abandoned

- **Physiology:** keep pressure 10–15 mmHg (12 mmHg more physiological). Pressure reduces venous return, raises afterload, causes **oliguria** (renal vein and parenchymal compression) and splints the diaphragm.

- CO_2 absorption plateaus at 30–60 minutes; higher with extraperitoneal insufflation. Hypercarbia peaks after extubation.
- **Gas embolism:** sudden collapse, millwheel murmur, end-tidal CO_2 rise then fall. Treat: stop insufflation, **left lateral, head-down**, 100% oxygen, aspirate via central line.

Comparison of open (RRP), laparoscopic (LRP) and robotic (RALP) radical prostatectomy

- **Background:** anatomical nerve-sparing open RRP (Walsh, 1982) was the standard. First LRP (Schuessler, 1997) was too slow; French centres standardised it in 2000, but laparoscopic suturing of the anastomosis remained a barrier. The robot (da Vinci, 2000) removed that barrier, and RALP is now dominant in the USA.

Parameter	Open RRP	LRP	RALP
Vision	Direct, loupes	2D, unstable	**3D, 10× magnified**, stable
Operating time	Shortest	Longest early	Longer than open; falls with experience
Instruments	Hands, haptic feedback	Rigid, fulcrum effect	**Wristed** (7 degrees of freedom); no haptics
Learning curve	Moderate	**Steepest** (suturing)	Shorter
Blood loss	300–1000 mL	Less (pneumoperitoneum tamponade)	Least; **transfusion 1–2%**
Hospital stay, catheter	Longer; catheter ~2 weeks	Shorter	Usually 1 day; catheter ~1 week
Pain, return to work	More	Less	Less; normal activity 3–4 weeks
Positive margins	Similar	Similar	Similar (~15%; ~9% if pT2)
Continence (1 year)	~93% (expert)	Similar	Similar; possibly earlier
Potency	Surgeon-dependent	Similar	Similar
Bladder neck stricture	Up to 16% (older series)	Lower	**<2%** (mucosa-to-mucosa)
Unique risks	Wound infection, hernia	Conversion, CO_2 effects	Steep Trendelenburg (eye, nerve), robot failure (<0.5%)
Cost	Lowest	Moderate	**Highest**

Evidence

- **Yaxley 2016 (randomised, one open vs one robotic surgeon):** equal continence, potency and early margins at 3 months; less blood loss and shorter stay with robot.
- **Coughlin 2018:** no difference in urinary or sexual function at 24 months.
- **LAP-01 (robotic vs laparoscopic):** earlier continence with robot.
- **Concept:** the **surgeon, not the platform**, decides outcome; margins depend mainly on tumour (Gleason ≥7, stage) and surgeon volume. The robot's advantages are recovery and wider reach (obese patients, prior surgery).
- **Also know:** transperitoneal vs extraperitoneal approach (no consistent difference); Retzius-sparing RALP (earlier continence, more margins); single-port (SP) robot; new robotic platforms; telesurgery.

Diagram to draw: Veress needle sites (umbilicus, Palmer point) and trocar layout for RALP.

Key points to remember:

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

I-3a. Non iodinated contrast materials

Type: SAQ (10 marks)

Introduction

- Used when iodinated contrast is risky (allergy, renal impairment) or with MRI (magnetic resonance imaging) and ultrasound.

1. Gadolinium-based contrast agents (GBCAs)

- **Mechanism:** paramagnetic gadolinium **shortens T1 relaxation** of nearby water → bright enhancement. Free gadolinium is toxic, so it is **chelated** (linear or macrocyclic; macrocyclic is more stable). Usual dose 0.1 mmol/kg; renal excretion.
- **NSF (nephrogenic systemic fibrosis):** skin and organ fibrosis, joint contractures, no proven treatment. Risk: eGFR (estimated glomerular filtration rate) <30 mL/min/1.73 m², dialysis, AKI (acute kidney injury).

ACR (American College of Radiology) group	Agents	NSF risk
I (linear)	Gadodiamide, gadopentetate, gadoversetamide	Highest
II	Gadobenate, gadoteridol, gadoterate, gadobutrol	**Little or none, even in ESRD (end-stage renal disease)**
III	Gadoxetate (liver-specific agent)	Few NSF data; recent ACR editions have moved some agents between groups, so check the current manual

- **Gadolinium deposition:** traces retained in brain (dentate nucleus) and bone, mainly with linear agents; no proven harm. Europe suspended most linear agents (2017).
- **Other:** allergic reactions rarer than with iodinated agents; not nephrotoxic at usual doses; avoid in pregnancy unless essential.
- **Uses:** renal mass, prostate mpMRI (multiparametric MRI), MR urography and angiography.

2. Ultrasound contrast (microbubbles)

- Gas core (perfluorocarbon or sulfur hexafluoride) in a lipid or albumin shell; strong harmonic echoes.
- Purely **intravascular**; gas exhaled by lungs. **No nephrotoxicity or radiation.** Rare cardiopulmonary reactions.
- **CEUS (contrast-enhanced ultrasound):** solid vs cystic renal mass (Bosniak), ablation monitoring, surveillance in CKD (chronic kidney disease). Intravesical use: **voiding urosonography for VUR (vesicoureteral reflux)**; ureteric patency after PCNL (percutaneous nephrolithotomy).

3. Other agents

- **CO_2 angiography:** negative contrast; not allergenic or nephrotoxic. Useful in CKD; **never above the diaphragm** (cerebral embolism).
- **Ferumoxytol** (iron oxide): MRI agent in renal failure. Lymphotropic iron nanoparticles help nodal staging (nodal metastases fail to darken).
- **Barium:** GI (gastrointestinal) studies only; contraindicated if perforation suspected.
- **Also know:** air as negative contrast (historical); radiotracers (^{99m}Tc, PET (positron emission tomography)) are not contrast agents; MRI safety (pacemakers, metal).

Diagram to draw: Chelated gadolinium (macrocyclic vs linear ring).

Key points to remember:

- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
- CO_2 angiography: never above diaphragm.

I-3b. Acetylcysteine.

Type: SAQ (10 marks)

Introduction

- **NAC (N-acetylcysteine):** acetylated form of the amino acid cysteine; a **thiol (sulfhydryl, –SH) compound**. Given orally, intravenously or by nebuliser.

Mechanisms of action

- **Glutathione precursor:** supplies cysteine, the rate-limiting substrate for **glutathione** synthesis.
- **Antioxidant:** the free –SH group scavenges reactive oxygen species.
- **Mucolytic:** –SH breaks **disulfide bonds** in mucoproteins, lowering mucus viscosity.
- **Thiol–disulfide exchange:** theoretically forms a soluble mixed disulfide with cystine.

Clinical uses

1. Paracetamol (acetaminophen) overdose: main indication

- Overdose saturates normal pathways; the toxic metabolite **NAPQI** (N-acetyl-p-benzoquinone imine) accumulates once glutathione is used up, causing hepatic necrosis.
- NAC restores glutathione. It is most effective **within 8 hours**, but still worth giving later.
- **IV (intravenous) 21-hour regimen:** 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h. Shorter two-bag regimens are now used in some countries.

2. Mucolytic: COPD (chronic obstructive pulmonary disease), cystic fibrosis.

Urological relevance

- **Contrast-induced AKI (acute kidney injury):** once widely used (early small trials, KDIGO (Kidney Disease: Improving Global Outcomes) 2012 suggestion). **PRESERVE (2018):** NAC gave no benefit over saline. It is **no longer recommended**; isotonic saline hydration is the only proven measure.
- **Cystinuria:** thiol action is theoretical; standard thiols are **tiopronin** and D-penicillamine, not NAC.
- **Cyclophosphamide cystitis: mesna** (also a thiol) is the agent used.
- **Mucus after enterocystoplasty or neobladder:** sometimes used orally or in irrigation; evidence weak.
- **Male infertility:** antioxidant for oxidative stress; limited evidence.

Adverse effects

- **Anaphylactoid reactions** with IV loading (flushing, rash, bronchospasm): slow the infusion, give antihistamine.
- Nausea, vomiting; unpleasant sulphur smell; bronchospasm when nebulised in asthma.

Diagram to draw: Paracetamol → NAPQI → glutathione conjugation, with NAC replenishing glutathione.

> **Key points to remember:**
>
> - Thiol; glutathione precursor, antioxidant, mucolytic.
> - Antidote for paracetamol overdose; best within 8 h.
> - PRESERVE 2018: no benefit in contrast AKI.
> - Cystinuria uses tiopronin, not NAC.

I-3c. Health related quality of life instruments (HRQOL) in urology.

Type: SAQ (10 marks)

Definition

- **HRQOL (health-related quality of life):** the parts of life affected by health: physical, emotional and social function and the patient's bother. It differs from a **symptom score**, which only counts symptoms.
- Must be patient-reported; doctors underestimate dysfunction.

Types of instruments

Type	Measures	Examples
Generic	Domains common to all diseases; allow comparison across diseases	**SF-36**, SF-12, **EQ-5D** (5 items + VAS (visual analogue scale); gives utilities for QALYs (quality-adjusted life-years)), Sickness Impact Profile
Cancer-specific	Cancer effects	**FACT-G** (Functional Assessment of Cancer Therapy–General), **EORTC QLQ-C30** (European Organisation for Research and Treatment of Cancer)
Disease-specific	Domains of one disease; more responsive	See below

Urology-specific instruments

- **Prostate cancer: EPIC (Expanded Prostate Cancer Index Composite)**, EPIC-26, UCLA (University of California, Los Angeles) Prostate Cancer Index, FACT-P.
- **Bladder cancer:** FACT-Bl, **Bladder Cancer Index**, EORTC QLQ-BLM30 and NMIBC-24, FACT-Vanderbilt Cystectomy Index.
- **Kidney cancer:** FKSI (FACT–Kidney Symptom Index).
- **LUTS (lower urinary tract symptoms): IPSS (International Prostate Symptom Score)** 7 items + 1 quality-of-life item; BPH Impact Index.
- **Incontinence and OAB (overactive bladder): ICIQ** (International Consultation on Incontinence Questionnaire), **OAB-q**, King's Health Questionnaire, I-QOL.
- **Sexual function: IIEF** (International Index of Erectile Function; 15 items), SHIM (5 items; no bother, so not true HRQOL), FSFI (Female Sexual Function Index).
- **Others:** NIH-CPSI (chronic prostatitis), O'Leary–Sant (interstitial cystitis), Wisconsin Stone QOL, USSQ (ureteric stent).
- **PROMIS (Patient-Reported Outcomes Measurement Information System):** free item banks, **computer-adaptive testing**, common T-score.

Psychometric properties (choosing a tool)

Property	Meaning	Threshold
Internal consistency	Items agree	Cronbach $\alpha > 0.7$ (>0.9 = redundant)
Test–retest reliability	Stable over time	$r > 0.7$
Content (face) validity	Right items present	Expert judgement
Criterion validity	Agrees with gold standard	$r \geq 0.7$

Property	Meaning	Threshold
Construct validity	Convergent and divergent	Correlations
Responsiveness	Detects real change	**MCID (minimal clinically important difference)**: AUA (American Urological Association) Symptom Index 3.1 points; otherwise ≥½ SD

- Validity is population-specific; translations need revalidation.
- Development: define domains → generate items → pilot → psychometric testing (Likert scales).

Diagram to draw: Pyramid: symptoms → function → bother → overall HRQOL.

> **Key points to remember:**
> - Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF).
> - SHIM lacks bother; I-QOL lacks symptoms.
> - Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.

I-3d. Distal Renal tubular acidosis.

Type: SAQ (10 marks)

Definition

- **Distal (type 1) RTA (renal tubular acidosis):** the collecting duct cannot secrete H^+ despite systemic acidosis. Result: **hypokalaemic, hyperchloraemic, normal anion gap metabolic acidosis** with **urine pH >6**. The only acidosis with alkaline urine.

Normal acidification and the defect

- **α-intercalated cells** secrete H^+ by apical H^+-ATPase; bicarbonate exits by basolateral Cl^-/HCO_3^- exchanger (AE1). Normally urine pH can fall to 4.5.
- In distal RTA, potassium is secreted in place of H^+ → hypokalaemia.

Causes

Inherited	Acquired (secondary)
SLC4A1 (AE1): mostly dominant	Sjögren syndrome, SLE (systemic lupus erythematosus)
ATP6V1B1, ATP6V0A4 (H^+-ATPase): recessive, **sensorineural deafness**	Obstructive uropathy, pyelonephritis, medullary sponge kidney
CA2: mixed proximal–distal	Hypercalciuria, hyperparathyroidism, amphotericin B

Why stones form (urological importance)

- **Stones in up to 70%**: **calcium phosphate** stones and **nephrocalcinosis**. Stones lead to diagnosis in up to half.
- **Profound hypocitraturia** (most important): acidosis increases proximal citrate reabsorption.
- **Hypercalciuria:** bone buffering releases calcium. **High urine pH** precipitates calcium phosphate.

Clinical features

- Stones, nephrocalcinosis, weakness (hypokalaemia), osteomalacia or rickets; children fail to thrive.
- **Incomplete distal RTA:** normal bicarbonate, but urine pH cannot fall below 5.5 after an acid load.

Diagnosis

- Low bicarbonate, high chloride, low potassium; urine pH >6 on a fasting morning sample.
- **UAG (urine anion gap) = $(Na^+ + K^+) - Cl^-$: positive** (low ammonium excretion) → renal cause; negative in diarrhoea.
- **Ammonium chloride loading** (0.1 g/kg orally; hourly urine pH): **failure to fall below 5.5** confirms. Furosemide–fludrocortisone test is better tolerated.
- 24-hour urine: low citrate, high calcium; ultrasound for nephrocalcinosis.

Treatment

- **Potassium citrate** (about 1–2 mEq/kg/day in divided doses): corrects acidosis, hypokalaemia and hypocitraturia, reduces calcium excretion and stone formation. Sodium alkali is less suitable because sodium raises urinary calcium.
- Treat the cause; manage stones endourologically.

- **Also know:** proximal (type 2) RTA (bicarbonate wasting, Fanconi, few stones); type 4 (hyperkalaemic, low urine pH; diabetes, obstruction); topiramate and acetazolamide mimic distal RTA.

Diagram to draw: α-intercalated cell with H^+-ATPase, AE1 and the stone mechanism.

Key points to remember:

- Urine pH >6 with acidosis; hypokalaemia; positive UAG.
- Calcium phosphate stones in up to 70%; hypocitraturia key.
- NH_4Cl test: pH fails to fall below 5.5.
- Potassium citrate is the treatment.

I-3e. PET scans in urology.

Type: SAQ (10 marks)

Principle

- **PET (positron emission tomography):** tracer emits a positron → annihilation with an electron → **two 511-keV photons at 180°** → coincidence detection. Fused with CT (computed tomography) or MRI.
- Quantitative (**SUV, standardised uptake value**); costly. Half-lives: ^{18}F 110 min, ^{68}Ga 68 min, ^{11}C 20 min (on-site cyclotron).

Tracers

Tracer	Mechanism	Main use
^{18}F-FDG (fluorodeoxyglucose)	Glucose analogue trapped after phosphorylation (Warburg effect)	Bladder, UTUC (upper tract urothelial carcinoma), penile, seminoma
PSMA (prostate-specific membrane antigen) ligands (^{68}Ga-PSMA-11, ^{18}F-DCFPyL, ^{18}F-PSMA-1007, rhPSMA-7.3)	Bind cell-surface PSMA	Prostate staging and recurrence
^{11}C/^{18}F-choline	Cell membrane synthesis	Prostate recurrence (older)
^{18}F-fluciclovine	Amino acid transport	Prostate recurrence
^{18}F-NaF (sodium fluoride)	Bone turnover	Bone metastases

Uses by organ

Prostate cancer

- FDG is poor (low glucose metabolism). PSMA has specificity >90%.
- **Primary staging (high risk): proPSMA (2020)**: PSMA PET/CT more accurate than CT plus bone scan (92% vs 65%). Nodal sensitivity only 30–40%, so a negative scan does not exclude micrometastases.
- **BCR (biochemical recurrence):** detection rises with PSA (prostate-specific antigen); about half positive below 0.5 ng/mL. PSMA beats fluciclovine (56% vs 26%). CONDOR: management changed in 63.9%.
- **Theranostics:** PSMA-positive mCRPC (metastatic castration-resistant prostate cancer) → **^{177}Lu-PSMA-617 (VISION 2021)**.
- NCCN (National Comprehensive Cancer Network): PSMA PET preferred; outcome benefit still unproven.

Testis

- **Seminoma residual mass >3 cm:** FDG-PET **≥6 weeks after chemotherapy**. Negative scan reliable (observe); positive → consider resection.
- **Not for NSGCT (non-seminomatous germ cell tumour):** teratoma is not FDG-avid.

Bladder and UTUC

- FDG for nodal and distant staging in MIBC (muscle-invasive bladder cancer): modest sensitivity, high specificity; urinary excretion hides the primary. Use when CT is equivocal.

Kidney

- FDG not routine (clear cell RCC (renal cell carcinoma) poorly avid). **^{89}Zr-girentuximab** (anti-CAIX, carbonic anhydrase IX) identifies clear cell RCC (ZIRCON).

Penis

- FDG confirms metastases in **palpable inguinal nodes**; poor for impalpable nodes.
- **Also know:** EMPIRE-1 (fluciclovine-guided salvage radiotherapy); false positives from inflammation; neuroendocrine cancer loses PSMA.

Diagram to draw: Positron annihilation with two 511-keV photons hitting a detector ring.

> **Key points to remember:**
> - proPSMA: PSMA PET 92% vs 65% accuracy.
> - Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
> - FDG poor in prostate and clear cell RCC.
> - VISION: ^{177}Lu-PSMA-617 in mCRPC.

I-3f. Complicated urinary tract infections.

Type: SAQ (10 marks)

Definition

- **Complicated UTI (urinary tract infection):** infection in a **structurally or functionally abnormal tract** or a **compromised host**, or with a resistant organism; more likely to fail or recur.
- **FDA (US Food and Drug Administration) 2018:** pyuria + pathogen in urine or blood + fever, chills, flank pain, with an abnormality or catheter.
- **Uncomplicated:** healthy non-pregnant woman, normal tract.

Complicating factors

Category	Examples
Obstruction	BOO (bladder outlet obstruction), PUJ (pelviureteric junction) obstruction, stricture, stone, tumour
Voiding dysfunction	Neurogenic bladder, high PVR (post-void residual), VUR (vesicoureteral reflux)
Foreign body / iatrogenic	Catheter, stent, nephrostomy, recent instrumentation, hospital-acquired
Anatomical	Diverticulum, fistula, cystocele
Host	**Male sex**, pregnancy, diabetes, immunosuppression, renal failure, transplant, elderly
Other	Symptoms >7 days; recent antibiotics

- Each causes stasis, biofilm or weak defences.

Organisms

- ***E. coli*** under half; more *Klebsiella*, *Proteus*, *Pseudomonas*, *Enterococcus*, *Candida*. Often **MDR (multidrug-resistant)** or ESBL (extended-spectrum β-lactamase) producers.

Evaluation

- **Urine culture mandatory before antibiotics**; blood cultures if febrile.
- Renal function, glucose, lactate.
- **Imaging:** ultrasound first; **CT (computed tomography)** if no response in 72 hours, stones, abscess or emphysematous infection suspected.

Treatment

- **Empirical IV (intravenous) therapy** by local resistance (EAU (European Association of Urology)): amoxicillin + aminoglycoside, 2nd-generation cephalosporin + aminoglycoside, or IV 3rd-generation cephalosporin. Avoid fluoroquinolones if used in the last 6 months or resistance >10%. Carbapenem or newer agents for ESBL (ceftolozane–tazobactam, ceftazidime–avibactam).
- **Source control is essential:** drain obstruction (**stent or nephrostomy, equally effective**), change catheter, drain abscess, treat stones later.
- Switch to oral when afebrile and stable; de-escalate by culture.
- **Duration:** usually 7–14 days (14 days in men if prostatitis not excluded); 7 days may suffice once afebrile ≥48 hours.
- **Sepsis:** antibiotics within 1 hour, fluids, vasopressors, ICU (intensive care unit).
- Correct the underlying cause, or infection recurs.

Special forms

- **Also know:** emphysematous pyelonephritis and cystitis (diabetics); pyonephrosis; renal and perinephric abscess; XGP (xanthogranulomatous pyelonephritis); CAUTI (catheter-associated UTI); funguria; unresolved bacteriuria (resistance, staghorn "critical mass").

Diagram to draw: Flowchart: suspect cUTI → culture → imaging → empirical IV → drain obstruction → oral step-down.

> **Key points to remember:**
> - Complicated = abnormal tract or compromised host; male sex alone qualifies.
> - Culture mandatory; often MDR.
> - Drain obstruction: stent = nephrostomy.
> - 7–14 days; oral step-down when afebrile.

Rapid Revision Sheet – Paper I

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

I-1. Describe anatomy of lymphatic drainage of genitourinary system, etiopathogenesis and management of Chyluria (LAQ)

- Right → left lymph flow; cisterna chyli at L1–L2.
- Chyluria = pyelolymphatic fistula, usually filarial (*W. bancrofti*).
- Ether test and triglycerides confirm; cystoscopy after fatty meal lateralises.
- Diet (MCT) + DEC → sclerotherapy → laparoscopic hilar lymphatic disconnection.

I-2. Discuss types of access, gases used in Laparoscopic urology and describe comparative analysis of open, Laparoscopic and Robotic Radical prostatectomy. (LAQ)

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

I-3a. Non iodinated contrast materials (SAQ)

- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
- CO_2 angiography: never above diaphragm.

I-3b. Acetylcysteine. (SAQ)

- Thiol; glutathione precursor, antioxidant, mucolytic.
- Antidote for paracetamol overdose; best within 8 h.
- PRESERVE 2018: no benefit in contrast AKI.
- Cystinuria uses tiopronin, not NAC.

I-3c. Health related quality of life instruments (HRQOL) in urology. (SAQ)

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF).
- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.

I-3d. Distal Renal tubular acidosis. (SAQ)

- Urine pH >6 with acidosis; hypokalaemia; positive UAG.
- Calcium phosphate stones in up to 70%; hypocitraturia key.
- NH_4Cl test: pH fails to fall below 5.5.
- Potassium citrate is the treatment.

I-3e. PET scans in urology. (SAQ)

- proPSMA: PSMA PET 92% vs 65% accuracy.
- Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
- FDG poor in prostate and clear cell RCC.
- VISION: ^{177}Lu-PSMA-617 in mCRPC.

I-3f. Complicated urinary tract infections. (SAQ)

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.